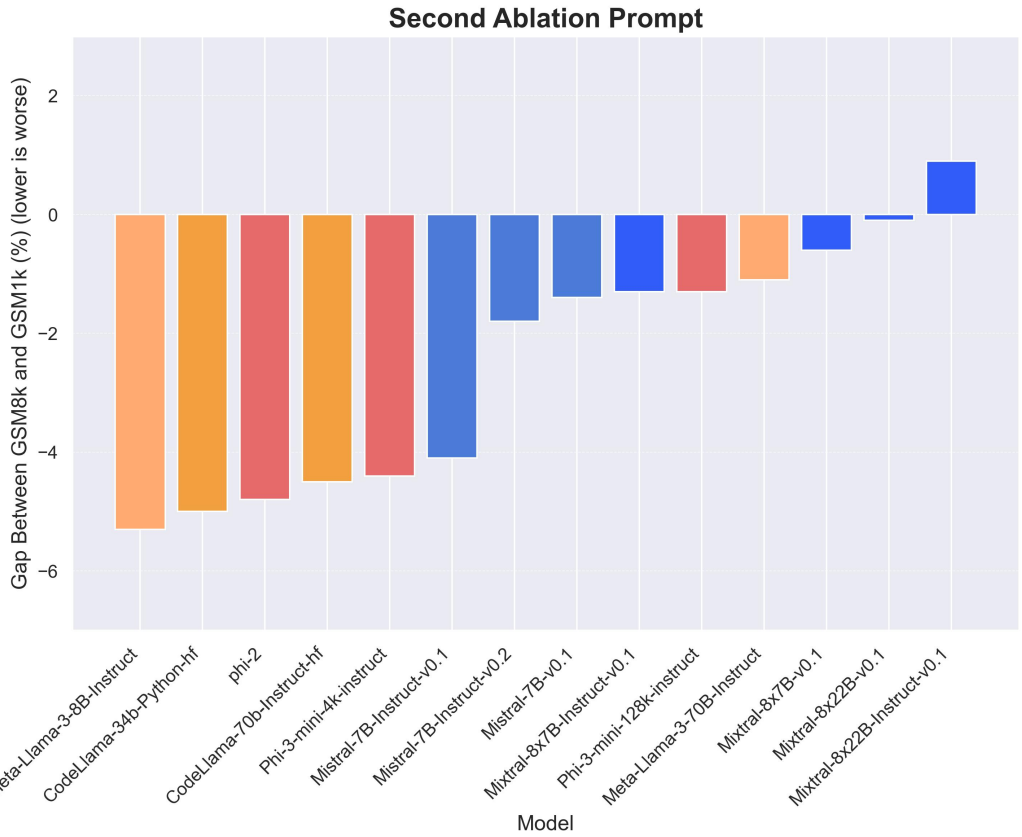

Figure 15: Models from the most overfit families arranged by their drop in performance between GSM8k and GSM1k (lower is worse) on the second ablation prompt with GSM8k examples in the alternative format.

L Ablations with Number of Fewshot Examples

As another ablation, we also investigate the impact of varying the number of GSM8k fewshot examples in the standard prompt format. We evaluate the models from the most overfit model families on the standard n-shot prompt, with n varying from 1 to 10, inclusive. As in the primary results, the n fewshot examples in the prompt are randomly selected from GSM8k train, and vary among the questions from the GSM1k/GSM8k test set. Note that the primary findings correspond to n=5. Our results are displayed in Figure 16. We notice that all models display a performance gap between GSM8k and GSM1k for almost all values of n, demonstrating a robustness to the overfitting result despite the variance from using different prompts.

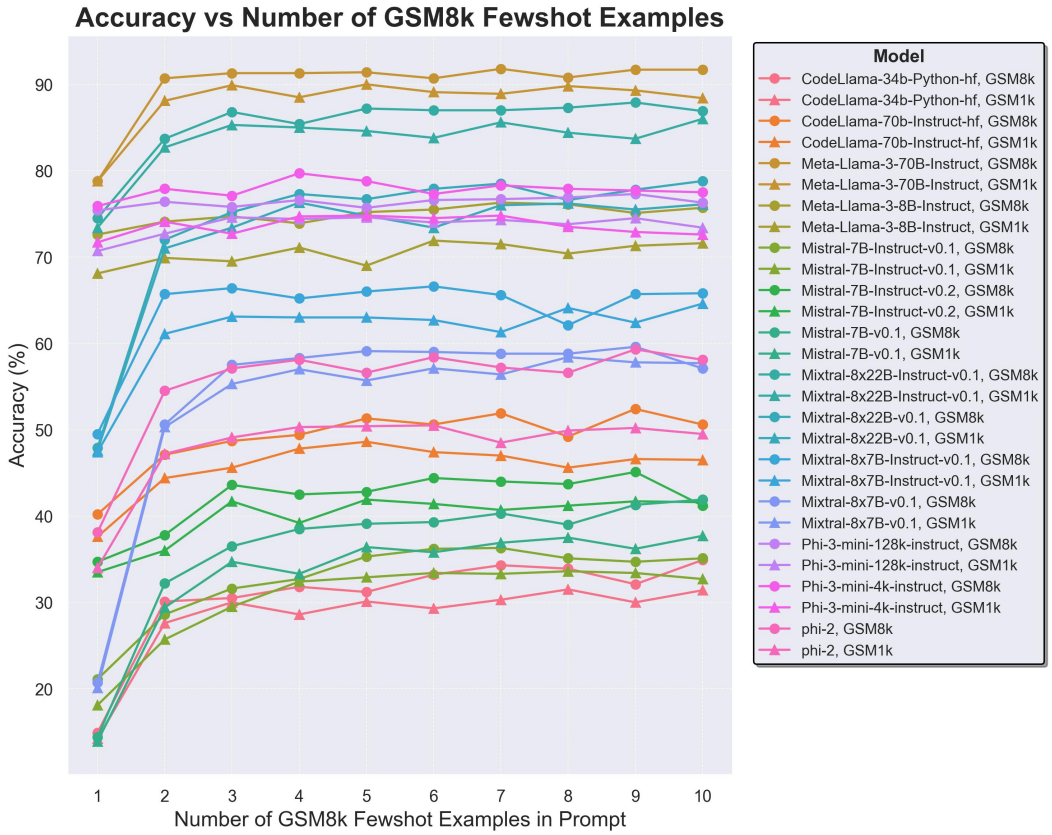

Figure 16: Performance on GSM8k and GSM1k relative to the number of GSM8k fewshot examples given in the standard prompt format, for models from the most overfit model families.

M List of Models Evaluated Using Default HuggingFace

We used vLLM to speed up model inference. A small number of models were not supported by vLLM at the time of initial evaluation, so we used the regular HuggingFace libraries to generate results. The list of these models is below.

- databricks/dbrx-base
- databricks/dbrx-instruct
- google/gemma-2b
- google/gemma-7b
- google/gemma-7b-it
- google/gemma-2b-it
- google/gemma-1.1-7b-it
- google/gemma-1.1-2b-it
- google/codegemma-7b
- google/codegemma-7b-it
- microsoft/Phi-3-mini-4k-instruct
- microsoft/Phi-3-mini-128k-instruct
- microsoft/Phi-3-medium-4k-instruct
- microsoft/Phi-3-medium-128k-instruct